PROGRAMME: BCA (Bachelor of Computer Applications) – DATA ANALYTICS

SEMESTER – V

Teaching-Learning & Evaluation Plan

Course Information:

Course Code: **23BCA5D01L** Course Title: **Data Warehousing and Data Mining Lab**

Credits Units: 01 Contact Hours: 30 L-T-P: 0-0-2

CA Weightage - 100 Pass Marks (CA) – 40 Special Examination Fees: NA

Course Facilitator : Dr.A.Kannagi, Associate Professor, School of CS & IT.

Programme Outcomes (POs)

At the end of the programme, students will be able to	
PO 1	**Computational Knowledge:** Understand and apply mathematical foundation, computing and domain knowledge for the conceptualization of computing models from defined problems.
PO 2	**Problem Analysis:** Ability to identify, critically analyze and formulate complex computing problems using fundamentals of computer science and application domains.
PO 3	**Design / Development of Solutions:** Ability to transform complex business scenarios and contemporary issues into problems, investigate, understand and propose integrated solutions using emerging technologies.
PO 4	**Conduct Investigations of Complex Computing Problems:** Ability to devise and conduct experiments, interpret data and provide well informed conclusions.
PO 5	**Modern Tool Usage:** Ability to select modern computing tools, skills and techniques necessary for innovative software solutions.
PO 6	**Professional Ethics:** Ability to apply and commit professional ethics and cyber regulations in a global economic environment.
PO 7	**Life-long Learning:** Recognize the need for and develop the ability to engage in continuous learning as a Computing professional.
PO 8	**Project Management:** Ability to understand management and computing principles with computing knowledge to manage projects in multidisciplinary environments.
PO 9	**Communication Efficacy:** Communicate effectively with the computing community as well as society by being able to comprehend effective documentations and presentations.
PO 10	**Societal & Environmental Concern:** Ability to recognize economic, environmental, social, health, legal, ethical issues involved in the use of computer technology and other consequential responsibilities relevant to professional practice.
PO11	**Individual & Team Work:** Ability to work as a member or leader in diverse teams in multidisciplinary environment.
PO12	**Innovation and Entrepreneurship**: Identify opportunities, entrepreneurship vision and use of innovative ideas to create value and wealth for the betterment of the individual and society.

Programme Specific Outcomes (PSOs)

PSO 01	Design, implement, populate and query the relational databases for operational data.
PSO 02	Import and evaluate a very large data sets to make business decisions.
PSO 03	Execute real time analytical methods on streaming data sets to react quickly to customer needs.
PSO 04	Mine data and carry out predictive modelling and analytics to support business decision making

Course Outcomes:

At the end of the course, students will be able to

Sl. No.	Course Outcome	Description	Bloom's Taxonomy Level
1.	CO1	Demonstrate knowledge on various Data acquisition process.	L3
2.	CO2	Illustrate data acquisition process from one data source to other target data source using data warehouse ETL tool.	L3
3.	CO3	Examine the weather, hospital, and banking datasets to discover knowledge for making future predictions using WEKA tool effectively.	L4
4.	CO4	Prove data mining problems with respect to documentation, visualization of hidden patterns and provide computing solutions for societal issues.	L5
5.	CO5	Develop solutions for complex computing problems by applying appropriate data mining algorithms to evaluate the accuracy and error measures using WEKA components.	L6

CO-PO/PSO Mapping: (3-Strong Correlation 2- Medium Correlation 1- Low Correlation)

Course Outcomes	Blooms Taxonomy Level	Programme Outcomes (PO)												Programme Specific Outcomes (PSO)			
		PO 01	PO 02	PO 03	PO 04	PO 05	PO 06	PO 07	PO 08	PO 09	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3	PSO 4
CO1	L3	3	3	3	3	3	1	1	-	2	-	-	2	3	2	2	2
CO2	L3	3	3	3	3	3	1	1	-	2	-	-	2	3	2	2	2
CO3	L4	3	3	3	3	3	1	1	-	2	-	-	2	3	3	2	2
CO4	L5	3	3	3	3	3	1	1	-	2	-	-	2	3	3	3	3
CO5	L6	3	3	3	3	3	1	1	-	2	-	-	2	3	3	3	3
	Average	3	3	3	3	3	1	1	-	2	-	-	2	3	2.6	2.4	2.4

Exp. No.	Name of the Experiment	Marks	CO
	PART –A **Creation of Active/Passive transformations using any open source Data Warehouse (Extract, Transform, Load) ETL Tool**		
1	Construct data acquisition process to extract, transform and load data from different databases.	25	CO1, CO2
2	Design and implement data acquisition process to perform a. Expression Transformation b. Joiner Transformation	25	CO1, CO2
3	Design and implement data acquisition process to perform • Aggregator Transformation	25	CO1, CO2
4	Implement mapping process to perform Filter transformation and Sorting transformation from one database to other database.	25	CO1, CO2
	PART -B **Working with Data Mining - WEKA tool**		
5	Creation on weather nominal and student results data sets in .arff and .csv formats	25	CO1, CO4
6	Perform data preprocessing steps on weather nominal and student information data sets as follows: a. Handling of missing values for categorical and nominal values. b. Selection of relevant attributes. c. Applying normalization techniques	25	CO1, CO4
7	Implement Association rule mining algorithm on preprocessed data set to find frequent item sets satisfying some threshold values.	25	CO1, CO4, CO5
8	Implement classification and prediction on processed data set using J48 and ID3 algorithms.	25	CO1, CO4
9	Evaluate the accuracy and error measures of a classifier or predictor using Experimenter WEKA component.	25	CO1, CO5
10	Minor Project Step 1: Creation of data set. Step 2: Apply preprocessing techniques on constructed data sets. Step 3: Implement appropriate data mining algorithms such as: • Apriori algorithm – to find frequent itemsets using various support and confidence levels • FP growth association mining ID3 decision tree classifier	25	CO1, CO5

Bloom's Taxonomy-Revised

LEVEL	DESCRIPTION	MEANING	ACTION VERBS
6	Creating	Can the student create a new product or POV?	Assemble, construct, create, change, combine, compose, design, develop, formulate, invent, modify, organize, propose, theorize, write
5	Evaluating	Can the student justify a stand or decision?	Appraise, agree, assess, argue, conclude, decide, defend, judge, prioritize, prove, rate, recommend, select, support, value
4	Analyzing	Can the student distinguish between different parts?	Contrast, compare, criticize, differentiate, discriminate, dissect, distinguish, examine, experiment, operate, question, simplify, test
3	Applying	Can the student use information in a new way?	Choose, demonstrate, dramatize, employ, illustrate, interpret, schedule, sketch, solve, use
2	Understanding	Can the student explain ideas and concepts?	Classify, describe, discuss, explain, identify, infer, locate, outline paraphrase, recognize, report, summarize, select, translate
1	Remembering	Can the student recall or remember information?	Define, duplicate, find, list, label, match, memorize, name, omit, recall, repeat, state, spell, tell

Signature of the Faculty **Signature of the HoD**